

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_4e32c8b8-725\agent-abf2b6d4c67c1f16c.jsonl`  
- SHA-256: `253c3377c6460fbd81b1697a451fc4a36130da901dc967edbea40614149706f1`  
- Source modified: `2026-06-19T17:19:47+00:00`  
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- project: `wf_4e32c8b8-725`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-19T17:18:10.917Z)

You are auditing the docs/ folder of the badminton-highlight-indexer repo to decide which docs can move to docs/archives/. Be rigorous and EVIDENCE-BASED ? the owner has run two doc-staleness sweeps and cares deeply about not archiving live work or breaking links.

REPO ARCHIVE POLICY (docs/archives/README.md) ? APPLY STRICTLY:

- Archive ONLY work in a clear TERMINAL state: DONE / COMPLETED / REJECTED / fully SUPERSEDED. "Never archive in-flight or pending work."
- KEEP-LIVE (do NOT archive) anything that is: living/active, in-progress, a not-yet-started but still-intended design/plan (owner-gated future work is LIVE, not terminal), a reference/navigation doc (CODE_MAP, README, HOW_RALLY_DETECTION_WORKS, EVALUATION), a template (PROJECT_DOC_TEMPLATE), or a roadmap/backlog/strategy doc still being tracked.
- Destinations inside archives/: past_projects/ (bounded completed project / tech-debt / starter effort) · research/ (completed or superseded research notes/plans) · decisions/ (ADRs) · checkpoints/ (dated state snapshots) · archives/ top-level (one-off audits/reports/launch records). Naming convention for a completed doc: <NAME>-COMPLETED-<date>.md.
- ALREADY-ARCHIVED COPIES EXIST for these (so the top-level file is likely a pointer-stub or a stale DUPLICATE to reduce, NOT a fresh move): docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md, docs/archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md, docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md. If your doc already has an archived copy, check whether the top-level is a stub (leave it) or a full duplicate (flag DEDUPE).
- CONTRADICTION TRAP: a doc may say "COMPLETED" yet still be cited as a LIVE source elsewhere (e.g. CLAUDE.md cites DEFERRED_HARDENING_PLAN.md #16 as the next platform slice). Before recommending ARCHIVE, grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and other top-level docs for the filename. If it is referenced as a live/next source, verdict = ASK_OWNER (note the contradiction), not ARCHIVE_NOW.

YOUR BATCH (read each from docs/<name>): GOLDEN_SET_IMPLEMENTATION.md, GOLDEN_SET_PHASE1_VIDEOS.md, GOLDEN_SET_VENDORING.md, GOLDEN_VIDEOS.md, GOLDEN_DATA_SHARING.md, GOLDEN_REGRESSION_FIXTURES.md

For EACH doc:

1. Read the status header + TL;DR/intro + (if present) the \$7/\$8 progress tracker + definition-of-done. You do NOT need to read every line of long docs ? sample enough to judge lifecycle state honestly.
2. Decide lifecycle and verdict per the strict policy. Remember: owner-gated / not-yet-started DESIGN docs are LIVE future work (KEEP_LIVE), NOT terminal. Only DONE/COMPLETED/REJECTED/fully-superseded is terminal.
3. Check for an already-archived copy (the policy lists three; also Glob docs/archives for the doc's stem). If the top-level is a thin pointer-stub ? KEEP_LIVE (it's intentional). If it's a FULL DUPLICATE of an archived COMPLETED copy ? DEDUPE_STUB (recommend reducing top-level to a stub or removing).
4. Cross-check "cited as live": Grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and the other top-level docs/ for this filename. If it's cited as a live/next/authoritative source, it

CANNOT be archived even if it says COMPLETED ? ASK_OWNER with the contradiction noted.

5. Inbound references: Grep the whole repo (docs/, backend/, CLAUDE.md, *.py) for the filename to count link-breakage risk; list up to 6 key referrers.

Return the structured object with one entry per doc in your batch. Be decisive: most reference/template/active/design docs are KEEP_LIVE; reserve ARCHIVE_NOW for genuinely terminal single-efforts/reports/completed projects, and ASK_OWNER for the contradictory/ambiguous ones.

2. assistant (2026-06-19T17:18:13.188Z)

I'll audit the six docs in my batch. Let me start by reading them and gathering cross-reference evidence.

3. user (2026-06-19T17:18:14.286Z)

ABLATION_LAYER.md
ANALYZERS_AND_RECONCILERS.md
BACKLOG.md
CODE_AUDIT_AND_TEST_HARDENING.md
CODE_MAP.md
COMMERCIALIZATION.md
DEFERRED_HARDENING_PLAN.md
DESKTOP_APP_PLAN.md
EVALUATION.md
EXECUTION_POLICY.md
EXPERIMENT_HARNESS.md
FIELD_VISIT_ANSWER_SHEET.md
FIELD_VISIT_PLAYBOOK.md
GOLDEN_DATA_SHARING.md
GOLDEN_REGRESSION_FIXTURES.md
GOLDEN_SET_IMPLEMENTATION.md
GOLDEN_SET_PHASE1_VIDEOS.md
GOLDEN_SET_VENDORING.md
GOLDEN_VIDEOS.md
HARDENING_LOOP_HANDOFF.md
HOSTING_PLAN.md
HOW_RALLY_DETECTION_WORKS.md
I18N_PLAN.md
KHELSUTRA_PRODUCT_OVERVIEW.md
MULTI_SIGNAL_FUSION_PLAN.md
NEXT_STEPS.md
OWNED_MODEL_IMPLEMENTATION_PLAN.md
PERSONALIZATION_PLAN.md
PER_VIDEO_OUTPUT_LAYOUT.md
PLATFORM_ARCHITECTURE.md
PMF_FIELD_SIGNALS.md
PMF_MARKET_RESEARCH.md
POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
PRODUCT_FIELD_ASSOCIATE_FLYER.md
PROJECT_DOC_TEMPLATE.md
PROMINENT_COURT_DETECTION.md
QUALITY_ITERATIONS.md
R1_MILESTONE_LAUNCH_REPORT.md
R2_EVAL_METRIC_DESIGN.md
RALLY_DETECTION_GAP_CLOSURE.md
RALLY_DETECTION_QUALITY_REPORT.md
RALLY_QUALITY_RESEARCH.md
README.md
REAL_FOOTAGE_VALIDATION_REPORT.md
ROADMAP.md
ROORA_LABELING_GUIDELINES.md
STORAGE_SHARING_MODEL.md
TRACKNET_WSL_SETUP.md
UI_ALPHA_EXECUTION_PLAN.md
UI_PROPOSAL.md

VIDEO_LOCALITY_MODEL.md
 VIDEO_RECORDING_GUIDELINES.md
 archives
 mockups
 ---ARCHIVES---
 CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
 COMMERCIAL_READINESS_REVIEW.md
 DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
 HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
 MONETIZATION_AUDIT.md
 README.md
 TEST_RUN_SUMMARY.md
 checkpoints
 decisions
 past_projects
 quality-iterations
 research

4. assistant (2026-06-19T17:18:16.182Z)

Let me read all six docs in my batch in parallel.

5. user (2026-06-19T17:18:17.145Z)

```

1      # Golden-Set Subproject ? Design & Implementation Overview
2
3      **Status:** In progress . **Date:** 2026-05-30
4      **Companion docs:** design rationale + vendor shortlist in
[GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md) (ADR-006); scoring mechanics in
[EVALUATION.md](EVALUATION.md).
5
6      This is the **implementation tracker** for turning the golden-set/vendor *design* into
shipped code. It's split into small, independently-reviewable PRs. The whole subproject is
**vendor-agnostic by construction** ? only one slice (GS-3, `HumanVendorProvider`) depends on
which vendor you pick offline; everything else can land first.
7
8      > **Update (2026-06-02):** Vendor selected for the **initial pilot ? Roora**
(Bengaluru). The **annotation guideline, intake-manifest template, and NDA** are authored in the
collector repo `docs/annotation/`, and the collector's `import-local` ingestion now covers the
**3 pilot sports** (badminton, pickleball, table tennis). On the indexer side, GS-1
(`file_provider.py`), GS-2 (`backend/eval/regression.py` + `run_regression.py`), and the GS-4
oracle scaffold (`automated_provider.py`) have landed; the remaining slice is the vendor-
dependent **GS-3 `HumanVendorProvider`**, parked until the Roora pilot confirms quality.
9
10     ---
11
12     ## Architecture in one picture
13
14     ```
15         video_ref ??? AnnotationProvider.submit()/poll()/fetch() ???
List[(start,end)]  ground truth
16                 ?           ?           ?           ?
17         FileProvider  HumanVendor  Automated(oracle)  ?
18         (GS-1)        (GS-3)        (GS-4)
backend.eval.metrics.score() ??? existing, unchanged
19
20         production pipeline ??? DB ??? backend.eval.harness.get_predictions() ???
List[(start,end)]  predictions
21
22                                     regress(video, tier) (GS-2)
compares the two
23     ```
24
25     **The keystone:** a provider's `fetch()` returns the **same `List[(start, end)]`** that
`eval.annotations.load_annotations()` already returns and `eval.metrics.score()` already

```

consumes. So one scorer grades all three of: product-vs-GT, vendor-vs-answer-key, and annotator-vs-annotator (IAA). No new scoring code ? only new *sources* of intervals.

```
26
27     ---
28
29     ## PR breakdown
30
31     | Slice | Branch | Scope | Depends on | Vendor-dependent? |
32     |---|---|---|---|---|
33     | **GS-1** | `feat/annotation-provider-registry` | `AnnotationProvider` ABC + registry +
`FileProvider` + tests | ? | No |
34     | **GS-2** | `feat/regression-runner` | `regress(video, tier, tolerance)` over existing
harness + `baseline.json` snapshot/compare + CLI | GS-1 | No |
35     | **GS-3** | `feat/human-vendor-provider` | `HumanVendorProvider` (submit/poll/fetch
against the chosen vendor's API or file-drop), DPA/secure-link handling | GS-1, **your vendor
pick** | **Yes** |
36     | **GS-4** | `feat/automated-oracle-provider` | `AutomatedProvider` fast-tier oracle ?
**interface stub** (injected labeler + fake; concrete model = backlog) | GS-1, GS-2 | No |
37     | **GS-5** | `feat/regression-triggers` | Wire triggers: CI gate, on-demand CLI,
scheduled run | GS-2, GS-4 | No |
38
```

Each row is one PR. GS-1 ? GS-2 ? GS-4 ? GS-5 can proceed **now**, independent of vendor selection. GS-3 is parked until you've chosen a vendor (then it's a thin adapter behind the GS-1 interface).

```
40
41     ---
42
43     ## Slice details
44
45     ### GS-1 ? Provider registry + FileProvider ? (this PR)
46     - `backend/annotations/base.py` ? `AnnotationProvider` ABC (`submit`/`poll`/`fetch` +
`annotate()` convenience) and `annotation_registry`, mirroring `backend/providers` and
`backend/indexer` registry patterns.
47     - `backend/annotations/file_provider.py` ? `FileProvider`: the "job" is a CSV that
already exists (formalizes today's manual/ShuttleSet flow); reuses
`eval.annotations.load_annotations`.
48     - Tests assert the registry contract **and** the keystone (provider output is directly
scorable by `metrics.score()`).
49     - **Design choice:** `fetch()`/`annotate()` raise loudly on a missing/incomplete job
rather than returning `[]`, so a broken label source can't silently masquerade as "0 rallies"
and skew a regression to green.
```

```
50
51     ### GS-2 ? Regression runner (next)
52     `regress(video, tier="fast", tolerance=0.05)`: get GT from
`annotation_registry.get(tier)`, read predictions via the existing `eval.harness`, score with
the existing `metrics`, compare against a per-video `baseline.json`, and flag a regression when
precision/recall drop beyond tolerance. Pure orchestration over existing parts + a small
baseline store. CLI: `python run_regression.py --video ... --tier file`.
```

```
53
54     ### GS-3 ? HumanVendorProvider (after your vendor pick)
55     A concrete provider that submits a (licensed/owned-only) video to the chosen vendor and
fetches the returned rally CSV. Carries the privacy controls from the design doc (secure time-
limited links, delete-on-delivery, DPA). Thin adapter ? the interface already exists from GS-1.
```

```
56
57     ### GS-4 ? AutomatedProvider (fast-tier oracle) ?? oracle rule
58     An independent strong labeler for the fast/CI tier. **Must be a different model lineage
than whatever production uses** (else shared blind spots ? false green). A fast-tier failure is
a *smoke alarm* that escalates to the GS-3 human tier, never an authoritative verdict on its
own.
```

```
59
60     ### GS-5 ? Regression triggers
61     Wire `regress()` into CI (pre-merge gate on pipeline changes), an on-demand CLI (the
"surprise regression"), and an optional scheduled run. Includes a deliberate re-snapshot path so
legitimate production improvements can rebaseline instead of raising false regressions.
```

```
62
```

```

63      ---
64
65      ## Conventions
66      - Branch per slice, one PR each, tests green before PR, full suite must stay green.
67      - New runtime deps: none through GS-2 (GS-3 may add a vendor SDK; GS-4 may reuse the
existing `google-genai` for a Gemini oracle).
68      - All ground-truth video stays licensed/owned ? see the privacy section of the
design doc.
69

```

6. user (2026-06-19T17:18:18.083Z)

```

1      # Phase-1 Golden-Set Video Collection Spec
2
3      Status: Design v1 . Date: 2026-06-02
4      Companion docs: [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md) (vendor process,
ADR-006) . [VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md) (capture spec the
user-facing guide derives from) . scoring in [EVALUATION.md](EVALUATION.md).
5
6      What footage to gather for the first round of golden-set creation so the trained
rally segmenter reaches good precision/recall ? and so the eval number actually predicts
production. Golden-set creation is iterative: expect multiple rounds, expanding toward
whatever axis round-1 metrics show is weakest.
7
8      ---
9
10     ## 1. The principle that dominates everything: match the deployment distribution
11
12     The CV literature is consistent: models trained on broadcast footage fail to
generalize to self-recorded amateur video ? broadcast uses a "straight, fixed camera with
minimum variability," while real non-elite footage has limited coverage, occlusions, and
inconsistent viewpoints ([team-sports
survey](https://link.springer.com/article/10.1007/s10462-024-10934-9), [unstructured-video event
detection](https://arxiv.org/pdf/2002.08097)). This is the same failure mode as ADR-004
(broadcast cameras track players, so player pixel-motion inverts during rallies).
13
14     Our product ingests users' own match recordings (a fixed phone/GoPro on a tripod).
Therefore:
15
16     - The golden set ? especially the held-out eval split ? must be dominated by
static, self-recorded, full-court footage that mirrors real users.
17     - ShuttleSet / broadcast footage is fine as training augmentation and as a
contrast set, but must be minimal in eval. Measuring precision/recall on broadcast yields a
number that does not predict production.
18
19     ---
20
21     ## 2. Primary recording profile (also the basis of the user-facing capture guide)
22
23     Grounded in the badminton-CV setup literature (camera ~2 m behind the baseline, ~4.5 m
elevated, ~30° down-tilt, whole court + headroom above the net ?
[source](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/)):
24
25     | Setting | Spec | Why |
26     |---|---|---|
27     | Mount | Fixed tripod; no pan/zoom during play | Avoids the ADR-004 camera-tracking
trap |
28     | Position | Behind one baseline/end, centered | Full-court, symmetric view |
29     | Height | 325 m elevated (~4.5 m ideal) | De-occludes the far player & net |
30     | Tilt | ~30° down | Optimal coverage, low keystoneing |
31     | Framing | Entire court + headroom above the net; players never leave frame |
Detection + court-relevance need it |
32     | Lighting | Even; no backlight/glare | Reduces blur & false motion |
33
34     Most of round 1 should use this profile; a minority deliberately varies it (§4) to probe

```

```

robustness.
35
36 ---
37
38 ## 3. Resolution & frame rate (per sport)
39
40 A controlled study: table-tennis ball detection was **22.5% @ 30 fps ? 93.2% @ 60 fps**
(120 fps ? 60 fps) ([Nature Sci Rep](https://www.nature.com/articles/s41598-023-42966-6));
OpenTTGames records at 120 fps. The ball/shuttle is the camera-invariant rally signal (our WASB
substrate, ADR-001), so fps is **not** a nicety for fast sports.
41
42 | Sport | Min resolution | Recommended fps | Note |
43 |---|---|---|---|
44 | Badminton | 1080p (720p floor) | 30 ok, **50/60 better** | Shuttle decelerates fast;
60 sharpens smashes |
45 | Pickleball | 1080p | **30?60** | Ball slower than TT; 30 likely fine, 60 safer |
46 | Table tennis | 1080p | **60 min (120 ideal)** | 30 fps collapses ball detection |
47
48 > These are **literature-backed starting points**; the published minimums get finalized
empirically against our golden set (VIDEO_RECORDING_GUIDELINES.md §3).
49
50 ---
51
52 ## 4. The Phase-1 diversity matrix
53
54 Cover the **high-impact axes** now; don't chase rare conditions in round 1. "Guards" =
whether the axis mostly protects **P**recision (not calling dead-time a rally) or **R**ecall
(not missing rallies).
55
56 | Dimension | Vary across? | Guards | Round-1 target |
57 |---|---|---|---|
58 | **Sport** | badminton, pickleball, table tennis | both | all 3 |
59 | **Format** | singles **and** doubles (badminton, pickleball) | R | both for those 2 |
60 | **Camera profile** | mostly primary-static; 1?2 side-elevated; 1?2 lower/phone-height;
**1?2 broadcast (contrast only)** | both | ~80% static / ~20% variants |
61 | **Venue / court color** | wood, green, blue synthetic | P | ?3 venues |
62 | **Lighting** | bright gym, dim club, mixed/backlit | both | ?3 conditions |
63 | **Background clutter** | empty hall vs people moving behind court | **P** (crowd
motion ? false positives) | both |
64 | **Players** | skill, attire colors, body types, L/R-handed, skin tone/age | R |
spread; don't reuse one pair |
65 | **Rally edge cases** | service faults, lets/replays, 1?2-shot & long rallies, net-
cords, coaching pauses, warm-up | boundary **P/R** | ensure present |
66 | **Resolution/fps ablation** | same match re-encoded 480/720/1080p; 30 vs 60 fps |
calibration | 1 matched set/sport |
67
68 ---
69
70 ## 5. Split it so the precision/recall number is honest (no train/eval leakage)
71
72 The #1 video-ML mistake: the **same match in train and eval inflates the score** ([data-
leakage guide](https://towardsdatascience.com/data-science-mistakes-to-avoid-data-
leakage-e447f88aae1c/)).
73
74 - **Split by match / recording session ? never by clip.** All clips from one match go
entirely to train *or* eval.
75 - **Hold out whole *conditions*** (a venue, a camera profile, a player pair the model
never saw) so eval measures generalization, not memorization.
76 - Reserve **~? of each sport as a deployment-like held-out eval set** (static, self-
recorded). Broadcast/ShuttleSet may enter train but should be **minimal in eval**.
77
78 > **Guardrails ? TODO (design with the user):** train and eval live in **separate
catalogs** today, kept apart by convention. We want hard guardrails ? permissions / explicit
approval / re-confirmation at ingest & export ? so a video can **never** silently land in both.
Tracked in BACKLOG.md; to be designed in a later session.

```

```

79
80      ---
81
82      ## 6. Concrete "round 1" shopping list (per sport)
83
84      Aligns with the 12?15-videos/sport scale plan; the 3-clip Roora calibration is a subset
of this.
85
86      - **8 static-primary** clips spanning venues / lighting / skill (?4 singles, 4 doubles
where applicable)
87      - **2 camera-variant** clips (side-elevated, lower height) ? robustness probe
88      - **1 resolution/fps ablation set** (one match @ 480/720/1080p, 30 & 60 fps)
89      - **1 broadcast contrast** clip (ShuttleSet) ? *train/contrast only*
90      - Each ~2?4 min, **deliberately including the edge cases** above ? **~12 clips/sport**;
hold out **4 whole matches** for eval.
91
92      ---
93
94      ## 7. Deliberately defer to later rounds
95
96      Outdoor courts, phone-handheld/jitter, extreme angles, 4K, unusual net/court variants,
crowd-heavy tournaments. Add these once round-1 precision/recall shows the weakest axis, then
expand toward it (find the quality "knee" ? VIDEO_RECORDING_GUIDELINES.md §3).
97
98      ---
99
100     ## 8. Two cautions specific to precision (the stated goal)
101
102     - **Backgrounds with people moving** behind the court are the biggest false-positive
driver. Include some so the model learns to reject them ? and make sure the annotation
guideline's "ignore all dead time" rule is airtight.
103     - **Warm-up / knock-up / practice swings** look like rallies. Capture some, and confirm
the vendor **excludes** them (the Roora guide now has an explicit rule + a how-to-spot-it tip).
Early precision is bounded more by **guideline / inter-annotator consistency (IAA)** than by raw
data volume.
104
105     ---
106
107     ## 9. Sources
108
109     - [Team-sports action-recognition survey ? Springer
2024](https://link.springer.com/article/10.1007/s10462-024-10934-9)
110     - [Unstructured-video event detection ? arXiv
2002.08097](https://arxiv.org/pdf/2002.08097)
111     - [Badminton service-height CV setup (camera 4.5 m / 30°) ?
PMC](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/)
112     - [VideoBadminton dataset ? arXiv 2403.12385](https://arxiv.org/html/2403.12385v1)
113     - [Table-tennis frame-rate vs detection accuracy ? Nature Sci
Rep](https://www.nature.com/articles/s41598-023-42966-6)
114     - [OpenTTGames (120 fps) dataset](https://lab.osai.ai/)
115     - [Action-spotting / segmentation survey ? arXiv
2505.03991](https://arxiv.org/html/2505.03991v2)
116     - [Data-leakage in train/test splits](https://towardsdatascience.com/data-science-
mistakes-to-avoid-data-leakage-e447f88aae1c/)
117

```

7. user (2026-06-19T17:18:18.594Z)

```

1      # Golden-Set Generation & Vendor Process (v1)
2
3      **Status:** Design v1 . **Date:** 2026-05-30 . **Vendor shortlist added 2026-05-30
($6)**
4      **Owner decisions captured:** licensed/owned video only . rally `[start,end)` depth .
tiered turnaround (fast automated + slow human) . vendor bake-off . **Bangalore/India-local
vendor preference**

```

5

6 This document defines how we produce ****golden-set ground truth**** (hand-verified rally boundaries) for the badminton highlights product, how we ****select annotation vendors****, and how that same machinery powers ****on-demand / surprise regression testing**** of the production tool.

7

8 > ****Companion docs:**** the scoring mechanics live in EVALUATION.md; the annotated-corpus rationale is [ADR-002](archives/decisions/DECISIONS.md); the decision record for ***this*** process is ****ADR-006****. Video-capture quality requirements (a dependency of trustworthy golden sets) are tracked separately in VIDEO_RECORDING_GUIDELINES.md. The ``regress()`` + ``baseline.json`` machinery in §5 is generalized into a ****variant A/B + promotion gate**** in EXPERIMENT_HARNESS.md.

9

10 > ****Update (2026-06-02):**** §6 shortlist done; the founder selected ****Roora**** (Bengaluru) for the ****initial paid pilot**** ? a ****single-vendor pilot first****, not the full bake-off below. Roora's site/capability/location are verified; ****quality is TBD via the pilot****. ****Step 0** (the annotation guideline) is authored ? collector repo ``docs/annotation/`` (``ROORA_BRIEF.md`` rally CSV schema + ``ANNOTATION_GUIDE.md`` illustrated guide + intake-manifest & NDA templates). Pilot scope is ****3 sports**** (badminton, pickleball, table tennis); the collector's ``import-local`` ingestion for all three is built. Still pending: the internal ****gold key**** and the vendor-dependent ****GS-3 `HumanVendorProvider`**** (GS-1/GS-2/GS-4 already landed ? see GOLDEN_SET_IMPLEMENTATION.md).

11

12 ---

13

14 ## 1. The keystone insight: one scorer, three jobs

15

16 A vendor's golden-set output is a list of rally ``[start,end)`` intervals ? ****structurally identical to a prediction**** from our pipeline. So the ***exact same*** deterministic scorer in ``backend/eval/metrics.py`` (``interval_iou`` ? ``match_intervals`` ? ``score``) grades all three of:

17

18 1. ****Product vs. ground truth**** ? the precision/recall/F1/IoU we already report (``EVALUATION.md``).

19 2. ****Vendor vs. our internal answer key**** ? the quality gate during vendor selection.

20 3. ****Annotator vs. annotator**** ? inter-annotator agreement (IAA), our proxy for "is this even a well-defined task?"

21

22 This symmetry is what makes the whole system small, auditable, and extensible. ****No new scoring code is needed**** ? only a new ***source*** of intervals (the vendor).

23

24 ---

25

26 ## 2. Why this also solves the privacy nuance

27

28 The monetization goal (hosted API) carries a "someone views the user's video" concern. This process sidesteps it ****by construction****:

29

30 - ****Golden sets are built only from footage we license or own**** ? starting with the founder's ****constantly-growing personal collection**** of match videos, plus ShuttleSet (ADR-003) and any purpose-recorded clips. ****No end-user uploads ever flow to a vendor.****

31 - Regression tests therefore run against a ****representative held-out corpus****, not literal user content.

32 - All video egress to any third party passes through a ****single provider boundary**** (§4), so the security controls (DPA, no-reuse, delete-on-delivery, access logging) live in exactly one place.

33

34 > ****Honest tradeoff:**** a licensed/owned corpus may not perfectly mirror real users' cameras, courts, and lighting. A green regression on our corpus is therefore ***strong evidence***, not a ***guarantee***, for production traffic. Revisit if field behavior diverges from corpus behavior ? and see VIDEO_RECORDING_GUIDELINES.md, which aims to shrink that gap by steering users toward capture conditions our corpus represents.

35

36 ---

37

38 ## 3. Selecting vendors ? the bake-off


```

39
40     ### Step 0 ? Write the annotation guideline (highest-leverage artifact)
41     Before contacting anyone, write a precise, example-rich definition of what a rally
42     is*:
43     - Start trigger: e.g. first serve motion vs. shuttle leaving the racket ? pick one,
44       show clips.
45     - End trigger: shuttle hits floor/net/out, or umpire call.
46     - Edge cases: lets/replays, service faults, between-point walking, coaching pauses,
47       mid-rally net-cord, broadcast replays and graphics.
48     Rally boundaries are genuinely subjective; ambiguity here is the #1 cause of bad
49     golden sets. This guideline belongs conceptually with the per-sport SportProfile, so it
50     generalizes to tennis/cricket later.
51
52     > Done (2026-06-02): authored as the Roora brief + illustrated guide in the
53     collector repo docs/annotation/. Defines the rally CSV schema and the three golden rules ?
54     ignore all dead time; include every rally (incl. service faults & lets); a shot = one
55     racket/paddle/bat contact ? plus an ending_reason taxonomy (winner / forced_error /
56     unforced_error / service_fault / let / other, shared across net-separated sports) and per-sport
57     detail incl. singles & doubles for badminton & pickleball.
58
59     ### Step 1 ? Build a small internal gold key
60     One trusted expert (initially: the founder) hand-annotates 5?10 licensed clips** to
61     near-perfect. This is the answer key vendors are graded against ? kept private during the bake-
62     off.
63
64     ### Step 2 ? Run the bake-off (parallel pilot)
65     Send the same pilot clips to 2?3 shortlisted vendors** (see §6, favoring
66     Bangalore/India), withholding the key. Grade every submission with our own harness:
67
68     | Dimension | How we measure it | Tool |
69     |---|---|---|
70     | Accuracy | vendor CSV vs. gold key ? P/R/F1, mIoU, boundary error | eval.score()
71     |
72     | Consistency | 2 annotators per vendor ? pairwise IoU (IAA) | interval_iou |
73     | Cost | quoted ? or $ per video-minute | vendor quote |
74     | Turnaround | actual vs. promised, per tier | observed |
75     | Ops friction | file-drop vs. API; compliance with our CSV format |
76     annotations.load_annotations() |
77     | Security | DPA signed; DPDP/ISO posture; delete-on-delivery | contract review |
78
79     ### Step 3 ? Decide on measured quality-per-dollar
80     The bake-off numbers are the decision ? not sales decks. Low IAA at a vendor means
81     either a weak vendor or a vague guideline (Step 0); investigate before blaming the vendor.
82
83     ### Step 4 ? Onboard the winner(s)
84     Sign DPA + the security controls in §4; codify the guideline as their spec; agree SLAs
85     for both tiers (§5).
86
87     ---
88
89     ## 4. The extensible architecture ? an AnnotationProvider registry
90
91     This codebase already uses pluggable registries (validators, segmenters, AI providers,
92     sports). The idiomatic move is the same shape for annotation sourcing:
93
94     ```python
95     class AnnotationProvider(ABC):
96         def submit(self, video_ref, guideline, sport) -> str:           # returns job_id
97             ...
98         def poll(self, job_id) -> str:                                   #
99         queued|running|done|failed
100         ...
101         def fetch(self, job_id) -> List[Interval]:                     # canonical
102         [(start,end), ...]

```

```

84         ...
85     ...
86
87     `fetch()` returns the same `List[Interval]` that `annotations.load_annotations()`
already produces, so everything downstream is unchanged.
88
89     Three implementations cover today plus both tiers:
90
91     | Provider | Role | Tier |
92     |---|---|---|
93     | `FileProvider` | Reads a pre-made CSV (formalizes today's manual / ShuttleSet flow) |
? |
94     | `HumanVendorProvider` | The bake-off winner; authoritative ground truth, corpus
growth, deep audits | Slow (days) |
95     | `AutomatedProvider` | An independent strong model labels rallies programmatically |
Fast (hours) |
96
97     The tiers are just two providers behind one interface, selectable per call.
98
99     ---
100
101     ## 5. The regression automation (the explicit ask)
102
103     > "given a video, get a golden set created and verify its precision and recall against
the production tool"
104
105     This becomes a thin wrapper over machinery that already exists:
106
107     ```python
108     def regress(video, tier="fast", tolerance=0.05):
109         gt = annotation_registry.get(tier).fetch(          # NEW: the provider
110             annotation_registry.get(tier).submit(video, GUIDELINE, sport))
111         predictions = process(video)                       # existing production
pipeline ? DB
112         s = eval.score(predictions, gt)                   # existing metrics.py,
unchanged
113         regressed = (s.recall < baseline[video].recall - tolerance or
114                     s.precision < baseline[video].precision - tolerance)
115         if regressed: alert(video, s, baseline[video])
116         return s
117     ```
118
119     Only one new component (the provider). `run_eval.py`, `harness.py`, and `metrics.py`
are reused verbatim. Snapshot a `baseline.json` per corpus video so "regression" is a measured
delta, not a vibe.
120
121     Triggers (all feed the same `regress()`): CI on pipeline changes · on-demand (the
"surprise" case) · scheduled cron · pre-release gate.
122
123     ### ?? The oracle problem (a real correctness trap)
124     The fast-tier `AutomatedProvider` must be a different model lineage than production.
If production becomes the offline ActionFormer segmenter (ADR-004) and the oracle is also
ActionFormer, they share blind spots ? the test shows green while both are wrong. Pairings that
stay honest:
125
126     - production = offline learned segmenter ? fast-tier oracle = paid Gemini full-video
(different lineage).
127     - production = Gemini hybrid ? fast-tier oracle = a different VLM or the slow human
tier.
128
129     Treat the fast tier as a smoke alarm, not ground truth: any fast-tier "regression"
escalates to the slow human tier for confirmation before it blocks a release.
130
131     ---
132

```

133 ## 6. Vendor shortlist (Bangalore/India-favoring)

134

135 > **Researched 2026-05-30 via a lean manual pass** (targeted web search + direct site/source verification ? NOT the heavy automated workflow, which failed). **Confidence is labelled per row.** **No pricing is listed because none could be verified** ? every vendor in this space quotes per-project, so ?/clip is an offline-conversation item (see §6.3). Treat single-source rows as **leads to verify on the call**, not vetted recommendations.

136

137 ### 6.1 Tier-1 candidates ? verified India presence + video capability (talk to these first)

138

139 | Vendor | Location (verified) | Video / temporal labeling | Notes for the call | Confidence |

140 |---|---|---|---|---|

141 | **iMerit** | **Bengaluru** office (A101 & A118, 43 Residency Rd, 560025; tech hub opened Oct 2022); also Kolkata HQ + US | Yes ? image/video/text/audio; explicit Sports & Biomechanics domain on their site | Largest & most enterprise-ready (5,500+ annotators); has a real **Bangalore** footprint for in-person coordination; ask for a sports/temporal-segmentation reference + a paid pilot. Higher-touch ? likely higher cost. | **High** |

142 | **TELUS International AI** (ex-**Playment**) | Playment was **Bangalore-founded** (2015); acquired by TELUS 2021, team retained | Yes ? built on Playment's 2D/3D image+video+LiDAR CV pipeline | Strong CV/video pedigree, Bangalore roots; but now a large global BPO ? confirm they'll take a **small startup pilot** and who the India delivery contact is. | **High** (origin) / **Med** (current India ops shape) |

143 | **Taskmonk** | **Bengaluru** | Yes ? "Video" listed as a labeling modality; platform **managed-service workforce** | Mid-size, India-local, does both tooling and service ? potentially the best **cost/agility** fit for a startup pilot. Verify temporal/event labeling specifically (their heritage is e-commerce catalog tagging). | **Med-High** |

144

145 ### 6.2 Tier-2 ? India video-annotation services, location/specialts to confirm on contact

146

147 | Vendor | Stated location | Note | Confidence |

148 |---|---|---|---|

149 | **Inforks** | India (Kerala-based per third parties) | Targets startups/SMBs; image/video/text/audio; positions on cost-efficiency ? good cheap-pilot candidate | **Med** |

150 | **Analytics** | India + USA (since 2018) | General data-annotation incl. video; widely listed | **Med** |

151 | **Objectways** | HQ Scottsdale AZ; **India delivery in Coimbatore** (not Bangalore) | Does video/LiDAR; India delivery but not Bangalore-local | **Med** |

152 | **Roora** (Roora ML Pvt Ltd) | **Bengaluru** ? HSR Layout (verified, `rooragrp.com`) | Image/video/text annotation; sports industry listed. **Selected for the initial paid pilot (2026-06-02)** ? existence/capability/location verified; **quality TBD via the pilot**. | **Verified** (capability) · pilot pending |

153 | **Tika Data** / **Zuru** | Claimed **Bangalore** (single-source listicle each) | Smaller/startup annotation shops ? unverified beyond one directory mention; verify they exist & do temporal video before investing time | **Low** |

154

155 ### 6.3 Explicitly NOT verified / out of scope

156 - **Pricing for everyone** ? all quote-only; do not assume. Get per-video-minute (or per-rally) quotes during the offline conversations.

157 - **Karya (Bengaluru)** ? confirmed local and reputable, but **speech/text/translation only, no video** ? **excluded** for our use case.

158 - **Sports/temporal track record** ? beyond iMerit's stated "Sports & Biomechanics" domain, vendors' specific **rally-boundary / temporal-event** experience is unconfirmed marketing copy; probe with the pilot (§3).

159 - **DPDP / DPA / ISO posture** ? not verified per-vendor; make it a checklist item in the bake-off (§3 Step 2 security row).

160

161 ### 6.4 Recommendation

162 For the **offline conversations**, start with **iMerit** (Bangalore-local, enterprise QA, sports domain) and **Taskmonk** (Bangalore-local, likely more startup-friendly cost/agility) as the two primary contacts, with **TELUS/Playment** as a third if you want a heavyweight CV pedigree. Run the §3 bake-off across whichever 2/3 agree to a paid pilot, and decide on measured quality-per-rupee. **Verify every "Confidence: Med/Low" claim on the call before committing.**

```

163
164     > **Update (2026-06-02):** the founder opted to start with **Roora** (Bangalore-local,
startup-friendly) as a **single-vendor pilot** rather than the iMerit/Taskmonk multi-vendor
bake-off above. iMerit / Taskmonk / TELUS remain the fallback shortlist if the Roora pilot
doesn't clear the §3/§4 quality bar.
165
166     > Sources (manual pass, 2026-05-30): iMerit Bengaluru hub ? imerit.net press /
PRNewswire (Oct 2022) + contact listings; Playment/TELUS ? BusinessWire & Inc42 (2021
acquisition, Bangalore-founded); Taskmonk & Karya locations ? company sites + AnalyticsIndiaMag
/ IndiaAI; others ? AnalyticsIndiaMag / Ziloservices / vendor sites. Pricing intentionally
omitted (unverifiable).
167
168     ---
169
170     ## 7. Suggested phasing
171
172     | Phase | Deliverable | Depends on |
173     |---|---|---|
174     | 0 | Annotation guideline + internal gold key (5?10 clips) | nothing ? can start now |
175     | 1 | Vendor bake-off ? pick winner(s) | §6 research + guideline |
176     | 2 | `AnnotationProvider` registry + `FileProvider` + `HumanVendorProvider` | winner
onboarded |
177     | 3 | `AutomatedProvider` (independent oracle) + `regress()` + baseline snapshots |
Phase 2 |
178     | 4 | Wire surprise triggers (CI / on-demand / scheduled) | Phase 3 |
179
180     Phase 0 is unblocked today and is the highest-leverage step regardless of vendor choice.
181
182     > **Status (2026-06-02):** Phase 0 guideline ? authored (collector `docs/annotation/`);
internal gold key still pending. Phase 1 is running as a **single-vendor Roora pilot** across
**3 sports** (badminton, pickleball, table tennis) rather than a multi-vendor bake-off. Phase
2's `FileProvider` (GS-1) and Phase 3's `regress()` + automated oracle (GS-2/GS-4) have already
landed; the main remaining slice is the **vendor-dependent `HumanVendorProvider`** (GS-3), with
trigger-wiring (GS-5) after it.
183
184     ---
185
186     ## 8. Open questions
187     1. **Guideline authorship** ? who owns the canonical rally definition, and is one
expert's key enough, or do we want 2-expert consensus for the answer key itself?
188     2. **Corpus representativeness** ? how well does the owned/licensed collection mirror
eventual user capture? (Ties directly to VIDEO_RECORDING_GUIDELINES.md.)
189     3. **Baseline drift** ? when production legitimately improves, how/when do we re-
snapshot `baseline.json` so old baselines don't raise false regressions?
190     4. **Fast-tier oracle cost** ? per-video Gemini cost at regression frequency (see
`archives/MONETIZATION_AUDIT.md` §4 for the per-video estimate).
191

```

8. user (2026-06-19T17:18:21.177Z)

```

1     # Golden-labeled video registry
2
3     **Purpose.** The single source of truth for every **golden** (human rally-boundary-
labeled) video ? what
4     it is, where the source proxy lives, how many rallies, and whether it's ingested into
the eval corpus.
5     Use this to **isolate the golden videos into a dedicated location** and to keep the
growing golden set
6     auditable. Updated whenever a new video is labeled + ingested.
7
8     > **This is the GROWING GOLDEN-finalized set** in the dual-eval-set discipline (see
9     > [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md), the 2026-06-16 capstone + `eval-dual-
set-regression`):
10    > a **RAW eval set** (currently-bad videos) + this **golden-finalized eval set**
(promoted when at-par) +

```

```

11     > a separate **golden TRAINING set**. Every quality launch must not regress on either
eval set.
12     > Privacy-safe, video-free *shipping* of these =
[GOLDEN_REGRESSION_FIXTURES.md](GOLDEN_REGRESSION_FIXTURES.md).
13
14     ## Contracts / conventions
15     - **Labels:** rally-annotator CSV
(`rally_number,start_time,end_time,ending_reason,sport,shots_count`,
16     decimal seconds) ? ingested by `backend/eval/gt_loader.py` (reads
`start_time`/`end_time`).
17     - **Alignment is mandatory before trusting any score.** The label timebase MUST match
the windowed proxy
18     (same fps + frame count). Verify with `ffprobe` (the `largetest_doubles` near-miss is
why) ? ideally
19     annotate the *exact* proxy the detector ran on (then `video_id` = MD5 matches and
alignment is free).
20     - **`video_id`** = MD5 of the proxy (`backend/utils/hashing.py::compute_video_id`); it
keys the WASB
21     trajectory cache, the Gemini reference rows, and the golden features.
22     - **Ingestion** = copy the trajectory + labels to durable paths, write
`output/<name>_golden_features.json`
23     (`gts` + candidates), append a row to `output/human_lovo_manifest.json`.
(`scratch/at_par.py` scores a
24     single video vs Gemini + golden; `backend/eval/rally_seg_eval.py` runs the whole
corpus.)
25     - **Guardrails:** owned footage only; **minors excluded**; labels never derived from a
paid LLM (Gemini is
26     inference/reference only). Commercial-clean.
27
28     ## Ingested golden corpus ? `output/human_lovo_manifest.json` (n = 15)
29
30     | # | name | sport | rallies | fps | source proxy | video_id | labeled |
31     |---|---|---|---|---|---|---|---|
32     | 1 | adarsh_avi_singles | badminton (S) | 96 | 59.94 | (original 6) | ? | pre-06-16 |
33     | 2 | kushagra_singles | badminton (S) | 32 | 59.94 | (original 6) | ? | pre-06-16 |
34     | 3 | largetest_doubles | badminton (D) | 49 | 29.97 | (original 6) | ? | pre-06-16 |
35     | 4 | gbaaddy | badminton | 48 | 59.94 | (original 6) | ? | pre-06-16 |
36     | 5 | testlarge_short | badminton | 6 | 29.97 | (original 6) | ? | pre-06-16 |
37     | 6 | mahadevpura_singles | badminton (S) | 31 | 29.97 | (original 6) | ? | pre-06-16 |
38     | 7 | mahadevpura_2 | badminton (multicourt) | 40 | 59.94 | `~/Roora Request - June
15/[Done] Video 3 ~/mahadevpura-2_proxy.mp4` | `038e0e0915a2` | 2026-06-16 |
39     | 8 | GX010128 | badminton (multicourt) | 52 | 59.94 | `~/Golden Label Videos Request -
June 15/[Done] Video 13 ~/GX010128.MP4` | `db84f3e65318` | 2026-06-16 |
40     | 9 | mahadevpura_1 | badminton (S, continuous) | 10 | 59.94 | `~/[Done] Video 14 -
Badminton/mahadevpura-1.MP4` | `38bef75d78cb` | 2026-06-17 |
41
42     ## Labeled, ingestion in progress (WASB trajectory extracting)
43
44     | name | sport | rallies | fps | source proxy | status |
45     |---|---|---|---|---|---|
46     | GX030094 | badminton | 41 | **29.97** | `~/Golden Label ?/[Done] Video 6 -
Badminton/GX030094_proxy.mp4` | **ingested** (in `human_lovo_manifest.json`) |
47     | Badminton BXH 2 | badminton (multicourt) | 44 | 59.94 | `~/[Done] Video 10 - Badminton
Multicourt/Badminton BXH 2_proxy.mp4` | **ingested** (in `human_lovo_manifest.json`) |
48
49     ## Source folder & not-yet-labeled
50     - **Annotation root:** `G:\My Drive\Sharing\Annotation Videos\Golden Label Videos
Request - June 15\`
51     (+ the earlier `Roora Request - June 15\`). `[Done]` prefix in a dir name = golden-
labeled.
52     - **Labeled badminton dirs:** Video 3 (=mahadevpura-2), 6 (=GX030094), 10 (=BXH 2), 13
(=GX010128, a
53     re-label of #8), 14 (=mahadevpura-1). **Video 13 is a duplicate of #8** (already
ingested).
54     - **Not yet labeled / other sports:** Video 1, 4, 7, 9, 12 (badminton, pending); Video

```

```

2, 8 (table tennis);
55     Video 5, 11 (pickleball). Badminton is the current focus (alpha customers);
TT/pickleball labeled later.
56
57     ## How to isolate the golden videos
58     Each row's **source proxy** column is the file to copy into a dedicated golden-videos
location. The labels
59     live beside the proxy (`<name>.rallies.csv`) and a durable copy is in
`output/<name>.rallies.csv`; the
60     trajectory is in `Collect/Trajectories/` (path in the manifest). Keep this table in sync
on every new ingest.
61

```

9. user (2026-06-19T17:18:22.121Z)

```

1     # Golden / GPU-Extractor Data Sharing ? GPU box ? dev sessions
2
3     **Status:** convention + plan (2026-06-15). Tracked in **#175**.
4     **Purpose:** a single cloud-synced location for the GPU-extracted golden artifacts
(trajecory CSVs, `_golden_features.json`, manifests, baseline snapshots) so the **GPU box**
and **CPU dev/agent sessions** share the same files ? without committing large data to git.
5
6     ## Why this exists
7     The rally-quality measurement is split across two machines by capability:
8     - **`backend/eval/fusion_golden.py`** (shuttle+person feature **extraction**) needs the
**videos**, the **trajectory CSVs**, the **`.rallies.csv` golden labels**, and a **GPU/WSL**
stack ? only the GPU box can run it.
9     - **`backend/eval/fusion_compare.py`** / **`backend/eval/ablation.py`** (the LOVO
**litmus**: ?F1, bootstrap CI, paired sign test) are **pure-CPU re-scoring** that read only the
small `_golden_features.json` ? any session (incl. a CPU dev container) can run them.
10
11     Today those artifacts sit in gitignored `output/` (per-machine) and scattered local
folders (`...\Annotation Setup\Collect\Trajectories`, `...\Golden Labelled`). Nothing syncs
them between machines, so a CPU session cannot see the GPU box's extraction. This wires that gap
with OneDrive.
12
13     ## Canonical shared folder
14     `%USERPROFILE%\OneDrive\rally-annotator\golden-data` ?
`C:\Users\avidu\OneDrive\rally-annotator\golden-data`
15
16     OneDrive is present + signed-in on both the GPU box and the dev machine under the same
account, so it auto-syncs. Layout (date/size-tagged so a re-extraction is a new folder, never an
overwrite):
17
18     ```
19     golden-data/
20         README.md
21         features/<corpus-tag>/          *_golden_features.json      (small ? the key shared
artifact)
22         trajectories/<corpus-tag>/      *_predict.csv / *_wasb.csv (medium ? reusable across
runs)
23         manifests/                     *_lovo_manifest.json
24         baselines/                     leaderboard / regression snapshots (or keep tracked in
eval_baselines/)
25     ```
26
27     `` = e.g. `2026-06-15-n6` (date + corpus size). **Do NOT** put raw videos
here (tens of GB) ? keep those local to the GPU box.
28
29     ## The GPU ? CPU workflow
30     1. **GPU box** (videos + CSVs + labels present):
31         ```
32         python -m backend.eval.fusion_golden --manifest <manifest> \
33         --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6"

```

```

34     ``
35     (or extract to local `output/` then copy the `*_golden_features.json` into the shared
folder).
36     2. OneDrive syncs the JSONs to the dev machine automatically (small files, seconds).
37     3. CPU dev/agent session points the analysis at the shared folder:
38     ``
39     python -m backend.eval.fusion_compare --features-dir "...golden-
data/features\2026-06-15-n6" --record
40     python -m backend.eval.ablation      --features-dir "...golden-
data/features\2026-06-15-n6" --granularity group
41     ``
42     ? real person-feature ?F1 / CI / sign test. Copy the JSONs into the repo's `output/`
to also light up the `skipif` golden-litmus tests
(`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`).
43
44     ## Optional code ergonomics (in #175, not required)
45     - A `RALLY_GOLDEN_DIR` env var (default `output/`) honored by `fusion_golden` /
`fusion_compare` / `ablation` so neither machine hardcodes the path. (The CLI `--out-dir` /
`--features-dir` flags already make this a convention; the env var is just ergonomics.)
46     - A tiny `scripts/sync_golden.py` to copy `output/*_golden_features.json` ? the shared
folder.
47
48     ## Guardrails
49     - The shared folder lives outside the repo ? never committed. Keep `output/`
gitignored and do not symlink the shared folder into a tracked path.
50     - The features derive from the Proprietary corpus ? keep the OneDrive folder
private (no public share links).
51
52     See also: `docs/NEXT_STEPS.md` (the "Next-session pickup" anchor),
`docs/MULTI_SIGNAL_FUSION_PLAN.md` (P3), `docs/TRACKNET_WSL_SETUP.md` (the GPU/WSL detector
setup), and docs/GOLDEN_REGRESSION_FIXTURES.md ? the COMMITTED, de-attributed regression
subset (synthetic + owner-cleared) that runs in-repo without the videos. This OneDrive
folder stays the full private corpus; the committed subset is the minimized/anonymized,
shuttle-only public tier (no person/audio, opaque surrogate ids).
53

```

10. user (2026-06-19T17:18:22.859Z)

```

1     # Golden Regression Fixtures ? video-free, non-attributable regression suite
2
3     > Status: `IN PROGRESS` ? owner-gated . Owner: Avi . Created: 2026-06-16 .
Last updated: 2026-06-16 . Legal: owner accepted the researched risk 2026-06-16
(conditions §3 stand; this is the owner's informed acceptance, not a retained-counsel
opinion)
4     > Lifecycle: `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE, per archive policy)
5     > Tracking anchors: §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
6     > Relation to existing docs: EXTENDS (does not supersede)
`docs/GOLDEN_DATA_SHARING.md` (#175); reuses the eval stack documented in `docs/EVALUATION.md` /
`docs/EXPERIMENT_HARNESS.md`; quality floor ties to `docs/QUALITY_ITERATIONS.md`.
7     > Honesty note: §3 (Legal) is researched general information, NOT legal advice ?
it gates on retained counsel (§8). §2/§4/§7 mark which claims are [verified] against code vs
[design] proposals.
8
9     ---
10
11     ## 0. TL;DR
12
13     Let a contributor on another machine with no video, GPU, WSL, or DB run the rally-
segmentation regression suite from committed structured files, so refactors and non-quality
PRs get a red/green side-effect signal. Two layers, built once and grown by accretion:
14
15     1. Mechanics ? freeze the deterministic pipeline at the post-detector trajectory
seam and replay it through the already-shipped eval stack (`experiment.py` / `metrics.py` /

```

```

>windowing.py` / `regression.py`), with two auto-discovered gates: a
**characterization/snapshot** gate (catches any behaviour change) and a **quality-floor** gate
(catches quality drops).
16     2. **Privacy/legal filter (SEALED-FIXTURES)** ? only **de-attributed, owner-source,
biometric-free** numbers are ever committed; richer/sensitive streams stay key-gated or out-of-
repo. Non-attributability comes from **deletion-at-source**, not encryption. ? "Minimized" means
**stream minimization** (shuttle-only, no person/audio/pose) ? **NOT** reduction of the shuttle
stream to ~5 scalars: the de-attributed per-frame `trajectory.csv` IS committed (see §4c + its
risk-acceptance re-confirmation note).
17
18     **The convergence that makes this clean:** the *privacy* answer (publish shuttle-only,
drop person/pose streams) and the *determinism* answer (freeze only the `CONTROL` path ? which
runs **no numpy execution**, keeping the numpy/BLAS learned scorer off the cross-machine gate)
point at the **same minimal public tier**. That coincidence is the spine of the design. (numpy
is still *imported* transitively via `experiment` and must be installed; the CONTROL replay just
never calls into it.)
19
20     **Two findings already actionable, independent of the rest:**
21     - ? **Live de-attribution bug** *(verified)*: real player/venue names are committed
today in `eval_baselines/heuristic_lovo_n6.json`,
`eval_baselines/gemini_wrapper_2026-06-10.json`, and `backend/eval/eval_partial_labels.py` (the
last also has a personal `C:/Users/avidu/...` path) ? and in git history. ? **Deliverable P0.**
22     - ? **Determinism trap** *(verified, corroborated by two independent red-teams)*: the
learned variants (`shuttle_only`, `fusion`) train a numpy/OpenBLAS `DYNAMIC_ARCH` logistic
regression whose decisions flip near the 0.5 threshold across CPUs ? would false-RED on a
contributor's machine. ? the cross-machine gate must be **CONTROL/static-only**.
23
24     ---
25
26     ## 1. Problem & goal
27
28     The rally-quality measurement is split across two machines by capability
(`docs/GOLDEN_DATA_SHARING.md`):
29     - `backend/eval/fusion_golden.py` **extracts** `_golden_features.json` ? needs the
**video + trajectory CSV + labels + GPU/WSL** (GPU box only).
30     - `backend/eval/fusion_compare.py` / `backend/eval/ablation.py` **re-score** those JSONs
on **pure CPU, no video** ? any machine.
31
32     The CPU re-scoring is *already* video-free, but the inputs live in gitignored `output/`
/ private OneDrive, so the regression tests that depend on them **skip everywhere**
(`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`). The
committed `eval_baselines/` baselines exist but `regression.py` still scores DB-stored
predictions (needs a pipeline run on the video).
33
34     **Goal:** commit a tiny, version-tagged, **numeric-only, non-attributable** per-fixture
corpus + auto-discovered gates so a clean `git clone` / CI gets a real red/green regression
signal with **zero video/GPU/DB**, while never leaking attributable or sensitive data.
35
36     ---
37
38     ## 2. Decisions locked
39
40     | # | Decision | Source | Implication |
41     |---|-----|-----|-----|
42     | D1 | **Corpus is owner-recorded / owned** | owner answer 2026-06-16 | Legal analysis
takes **Fork A** (clean): no third-party copyright/ToS infringement; concern is trade-
secret/moat + privacy. Strongest non-attributability (no public reference to correlate against).
|
43     | D2 | **Repo private now, may open-source later** | owner answer 2026-06-16 | Design
for **public-grade hygiene from day one** ? git history is permanent; nothing scrubable-later. |
44     | D3 | **Adults, consent not formalized** | owner answer 2026-06-16 | **Person/pose/gait
streams stay OUT of the public tier** (DPDP/GDPR derived-personal-data risk without consent).
Public tier = **shuttle + abstract intervals + non-biometric features only**. |
45     | D4 | **Freeze seam = post-detector trajectory** *(design; SHIPPED differs ? see ?)* |
workflow-1 synthesis | widest deterministic surface; reuse the shipped eval stack. ?

```


****Correction (audit 2026-06-16):**** the shipped freeze ****commits** the de-attributed per-frame `(Frame,Visibility,X,Y)` `trajectory.csv` ****** (shuttle-only, time-origin reset) ? it is ****NOT**** minimized to ~5 aggregate scalars. See §4c (and its risk-acceptance re-confirmation note). |

46 | D5 | ****Cross-machine gate is CONTROL/static-only**** *(design)* | both red-teams [verified cause] | Learned variants (numpy/BLAS) are ****owner-box/Tier-1 only****, asserted with tolerance, never as a cross-machine byte-snapshot. |

47 | D6 | ****Non-attributability = deletion-at-source, encryption = public-only barrier**** *(design)* | workflow-2 synthesis | A contributor who runs the tests necessarily reads plaintext+key (DRM/analog-hole). So we ****delete**** attribution, not hide it. Encryption only stops the no-key public. |

48 | D7 | ****Two tiers**** *(design)* | workflow-2 synthesis | ****Tier-0**** public/no-key (shuttle-only, de-attributed). ****Tier-1**** key-gated/out-of-repo (person/audio, full corpus), ****skips cleanly when absent****. |

49

50 ---

51

52 **## 3. Legal foundation (researched ? gates on counsel, §8)**

53

54 ****Core verdict (settled across IN/US/EU):**** the derived numbers ? per-frame shuttle (x,y), rally start/end intervals, audio/court/aggregate-person feature values ? ****do NOT inherit** the source video's copyright. ****** They are uncopyrightable *facts/measurements*, not authored expression, and not a §101 "derivative work."

55

56 | Jurisdiction | Copyright of the data | Database right | Contract / ToS | Net |

57 |---|---|---|---|---|

58 | ****India**** (primary) | Not copyrightable as facts (*EBC v. D.B. Modak*); but compilations get thin protection on a low "skill/labour" bar (*Burlington*, *OLX v. Padawan*) ? only bites if ****lifted from a 3rd-party dataset****, not self-derived | ****None**** (no sui-generis right) ? a real permissiveness edge | Anti-scraping ToS (untested in court) + ****IT Act s.43/s.66**** civil/criminal exposure for 3rd-party ingestion | ****Owned-source + minors-excluded + no person/pose = clean**** |

59 | ****US**** | Not copyrightable (*Feist*; §102(b); *NBA v. Motorola*) | None (*Feist* killed sweat-of-the-brow) | ****Dominant risk.**** ToS enforceable where copyright/CFAA fail (*ProCD*, *hiQ* ? injunction + ****forced deletion****). Remedy for redistribution is takedown, not nominal damages | Publishable as facts ****iff**** owned/permissioned source + no barring ToS + no person/biometric/minor stream |

60 | ****EU / UK**** | Not copyrightable (*Football Dataco v. Yahoo*) | ****Genuinely contestable.**** *Football Dataco v. Sportradar*: recording pre-existing on-court facts = ****obtaining****, not "creating" ? our "we created it" premise is ****overconfident****; harm test = *CV-Online Latvia* | Ryanair v. PR Aviation: ToS can ban extraction/redistribution even with no IP right | Get EU+UK advice before redistributing numeric streams at scale |

61

62 ****Provenance is the decisive fork.**** ****Fork A (owner-recorded, unpublished)**** ? the only path safe to commit publicly; collapses to a trade-secret *choice* (Indian breach-of-confidence / Contract Act 1872 / IT Act s.72 ? ****not**** US DTSA) + privacy. ****Fork B** (broadcast/YouTube/scraped) ? do ****not**** commit; the extraction act needs a license or now-****contested**** US fair use (*Thomson Reuters v. Ross* 2025; USCO May-2025), and unlawful acquisition independently sinks it (*Bartz v. Anthropic*, ~\$1.5B settlement). Caveat: owner footage ****shot at a third party's event**** (tournament/club) carries ticket/venue/organisier/broadcaster terms ? treat as Fork B until cleared.

63

64 ****Publish-by-stream (privacy):****

65 - ? ****Publish-OK:**** shuttle (x,y)+Visibility (inanimate object), abstract rally/GT ****intervals**** (identity-stripped), non-biometric audio/court/aggregate-person features detached from identity.

66 - ?? ****Conditional (effectively never in v1):**** person bounding boxes that *persistently single out a player* ? flips to GDPR Art.9 the moment it identifies. This architecture is prone to that flip ? treat per-player tracking as a tripwire.

67 - ? ****Never publish:**** pose/skeleton/****gait**** (re-identifies at 80?95% even with faces blurred ? *Weiss et al. 2025*), face crops, any named-player?movement mapping. Matches the existing "gait/biometrics strictly future (GDPR Art.9)" guardrail.

68 - ? ****Minors flip everything:**** DPDP "child" = under 18; s.9(3) ****prohibits behavioural monitoring**** (the Rule-12 exemptions don't cover this project). A persistent per-player signature on a minor is, conservatively, prohibited. ****Exclude minors**** from any persisted stream beyond inanimate shuttle/interval data.

69

70 **Non-attributability is relative, not absolute.** A per-video trajectory is a
fingerprint (de Montjoye: 4 points ? 95% re-ID). It's only non-attributable because the **Fork-A
source stays unpublished** (no public reference to correlate against). Per *EDPS v. SRB* (CJEU
C-413/23 P, 2025) it **remains personal data to the holder** who can re-run the detector. So:
conditional, revocable (publish the source and the link snaps back), forward-looking (re-assess
as the public-video environment grows). Never market it as "anonymous."

71

72 *Key authorities:* Feist 499 U.S. 340; NBA v. Motorola 105 F.3d 841; 17 U.S.C.
§101/§102(b); Sega v. Accolade 977 F.2d 1510; Football Dataco v. Yahoo (C-604/10) & v.
Sportradar (C-173/11); BHB v. William Hill (C-203/02); CV-Online Latvia (C-762/19); Ryanair v.
PR Aviation (C-30/14); Thomson Reuters v. Ross (D. Del. 2025); EBC v. D.B. Modak; DPDP Act 2023
s.3/s.9 + Rules 2025; EDPS v. SRB (C-413/23 P). Full URLs in the research artifact (§10).

73

74 ---

75

76 ## 4. Architecture (design)

77

78 ### 4a. Freeze seam + reuse map

79 Freeze at the **trajectory CSV ? CONTROL-path replay**. One shared helper
`backend/eval/golden\_fixtures.py::replay\_fixture()` is the *single* code path used by both the
freeze tool and both gates. As **shipped**, that path is (per the module docstring):
80 `parse_trajectory_csv ? distill_local.build_candidates(proposal=(vt,mg,mwd)) ?
VideoCandidates(5 shuttle cols only) ? experiment.predict(CONTROL) [keep all ? merge_overlaps] ?
metrics.score`.

81 Reuses **verbatim**: `metrics.score/interval\_iou`,
`experiment.predict/CONTROL/VideoCandidates`, `distill\_local.build\_candidates`,
`tracknet\_runner.parse\_trajectory\_csv`/`trajectory\_to\_action\_windows`, `rally\_gate` net-
crossings (inside `build\_candidates`), `gt\_loader.load\_gt\_intervals` (strict),
`regression.gt\_hash`, `eval\_baselines/` committed-baseline precedent.
82 > ? **Correction (audit 2026-06-16)**: the replay does **NOT** import `windowing.py` ?
there is no `windowing.build\_windows(preset)` / `WindowingPreset` / "default SERVED" call. The
windowing params travel as a **raw 3-tuple `(vt, mg, mwd)`** pinned in each fixture's
`provenance.json` and passed straight to `build\_candidates`; the harness deliberately avoids the
named `windowing.SERVED` constant so a future SERVED retune can't silently churn fixtures.
(`segmentation\_metrics`, `experiment.compare/Variant.spec\_hash`, `regression.save\_baseline`,
`eval\_stats` are **not** on the M1 replay path either ? the quality-floor gate M4 will add the
baseline/stats pieces.)

83

84 ### 4b. Tiered model (SEALED-FIXTURES)

85 | Tier | Where | Key? | Contents | Test behaviour |

86 |---|---|---|---|---|

87 | **Tier-0** | `eval\_baselines/fixtures/` (tracked) | none | opaque surrogate slug,
fps/frame_width (retained, NOT quantized), the **de-attributed per-frame `trajectory.csv`**
`(Frame,Visibility,X,Y)` (shuttle-only, time-origin reset), the 5 shuttle feature columns inside
`expected.json`, strict-slug `label`, relative `gts`. **No person/audio/pose.** ? NOT minimized
to 5 scalars ? see §4c. | **Unconditional** ? runs on every PR/fork; turns today's skipped
litmus into a real public gate |

88 | **Tier-1** | OneDrive (default) **or** SOPS-encrypted in-repo | token/age key | full
corpus **with** person/audio; authoritative ?F1/CI/sign-test + exact Gate-A litmus | **Skips
cleanly** when key/data absent (generalize `\_golden\_present()` ? `\_tier1\_present()`) |

89

90 A fresh no-key clone is always **green**; coverage never depends on a secret.

91

92 ### 4c. Anti-attribution measures (Tier-0)

93

94 > ? **RISK-ACCEPTANCE RE-CONFIRMATION REQUIRED (audit 2026-06-16).** An earlier draft of
this doc claimed Tier-0 commits "only ~5 aggregate shuttle scalars" and "***never** the per-frame
CSV". **That is not what ships.** The shipped freeze (`freeze\_real\_fixture`) **commits the FULL
per-frame `(Frame,Visibility,X,Y)` `trajectory.csv`** ? de-attributed (time-origin reset +
opaque random slug + metadata strip), but **not minimized to 5 scalars**. The per-frame stream
is the de Montjoye fingerprint (4 points ? 95% re-ID against a published reference). The owner's
2026-06-16 risk acceptance (§7 P2) was recorded against the inaccurate "5 scalars" basis and
MUST be RE-CONFIRMED on this corrected basis before any **real** Tier-0 fixture is committed
(P3). Synthetic fixtures (current state) carry zero provenance risk regardless.

95

```
96 - **Opaque RANDOM surrogate id** (`os.urandom` + a private surrogate?source map kept
off-git) ? *not* the MD5 `video_id`, and *not* HMAC(salt, video_id?name) (red-team: brute-
forceable over a tiny known roster if the salt leaks). Never persist the source `name`/filename.
97 - **Strict-slug `label` + dropped `source_note`** ? the operator `label` is validated
against `^[a-z0-9][a-z0-9_\-]*$` (a free-text identifier like a player/event name is refused),
and operator free-text `source_note` is **accepted-but-NOT-persisted** (never reaches the
committed `provenance.json`) ? both close the free-text identifier leak `_scan_forbidden` could
not police (audit fix #1).
98 - **Metadata strip** ? no filename, video_id, match/event/player identity, capture date.
99 - **Time-origin reset** ? subtract per-fixture `t0` (6dp-rounded) from all
start/end/gts. **Metric-invariant within `_FLOAT_TOL`** (`metrics.interval_iou` and start/end-
error are pure differences): the genuine (raw) reset-vs-no-reset score delta sits at the float-
error floor (~3.3e-7); bit-identical for whole-second offsets. See §4c-note below and §6.
100 - **Per-frame trajectory IS committed (de-attributed), NOT minimized to scalars** ?
Tier-0 commits the de-attributed per-frame `trajectory.csv` (shuttle stream only). It is **not**
reduced to ~5 aggregate scalars; the 5 shuttle *feature* columns appear (computed) inside
`expected.json`, but the raw per-frame `(Frame,Visibility,X,Y)` stream is part of the committed
fixture. (This is the corrected basis the ? note above gates on.)
101 - **Stream minimization (streams, not resolution)** ? Tier-0 keeps the shuttle block
only; person/audio excluded; pose/gait/face have **no field in the schema by design** (positive
`_scan_forbidden` assertion on the written bytes).
102 - **fps/frame_width are RETAINED as scale constants** ? they are NOT quantized today
(quantization is a deferred P3 option, see §7 ?). Reconcile any "quantize fps/frame_width"
wording elsewhere to this.
103 - **Provenance hard-fail gate** ? the de-attribution pass refuses to emit a public
fixture for any record not tagged owner-recorded(Fork-A) + minors-excluded + non-blank-license,
and refuses a negative reset label (timeline predates the trajectory). Writes go to a temp dir
and are promoted only after the privacy scan passes, so a refused freeze leaves no partial
fixture (audit fixes #3a/#3b).
104
105 ### 4d. The two gates
106 - **Characterization/snapshot** ? replay fixture, **parse-compare** JSON (never byte-
compare ? CRLF/float-format safe; `core.autocrlf=true` here), exact on int/structural fields,
`approx(abs=1e-6)` on floats; stamp + check windowing-preset/label/gt fingerprints
(incommensurable ? loud fail). Catches any behaviour change even at constant F1. **CONTROL path
only** across machines.
107 - **Quality-floor** ? score vs committed labels; assert per-video + mean F1 ? floor in
`eval_baselines/regression_baseline.json` (created here; carries a **windowing fingerprint** so
a served-default change is flagged incommensurable, not silently compared); emit `eval_stats`
paired-delta-CI + sign test (NUMBERS INFORM); add a **metrics-vs-base-branch** paired delta.
Synthetic-tier floors measure **correctness, not field quality** (documented).
108
109 ### 4e. Serialization & optional protobuf
110 JSON with `sort_keys=True` + pre-quantized floats + LF is sufficient. **Protobuf is
optional (O1)** ? adopt only if cross-OS byte-stable snapshots are wanted (proto3
`rally.fixture.v1`, codegen + `protoc && git diff --exit-code` guard, `*.pb` binary -text`).
Protobuf is **encoding, not secrecy** (trivially `--decode_raw`).
111
112 ### 4f. Encryption & keys (Tier-1 only)
113 **Default = out-of-repo + token** (existing OneDrive `RALLY_GOLDEN_DIR` seam ? zero
sensitive bytes in git history, sidesteps the legal "redistribution" act). **Alternative = SOPS
+ age** (per-recipient, revocable, value-level diffs) for Fork-A-clean-but-private data wanting
in-repo versioning. Rejected: git-crypt (whole-repo re-key), git-lfs (storage ? secrecy). CI
secret exposed **only to the Tier-1 job on trusted branches, never to fork-PR runners**. The
**HMAC/surrogate salt** (if used) and the surrogate?source map live with the private corpus,
rotated per release. Document explicitly that Tier-1 crypto defends **only** the no-key public.
114
115 ---
116
117 ## 5. Threat model (who's protected vs not)
118
119 | Actor | Protected against | NOT protected (by design / residual) |
120 |---|---|---|
121 | **TM1 no-key public** | map fixture?source/match/player; decrypt Tier-1; confirm a
```

```

source even holding the candidate file | read Tier-0 numbers + run shuttle-only harness
(intended; useless-in-isolation) |
122 | **TM2 keyed contributor** | recover filename/video_id/match/face/pose/gait ? *never
written* (survives the analog hole) | read every value the suite reads (unpreventable; not
pretended otherwise) |
123 | **TM3 CI runner** | leaked log can't leak attribution it never had | runs the suite
(intended); Tier-1 secret never on fork runners |
124 | **TM4 malicious keyed insider** | correlation needs a public reference; Fork-A source
is unpublished ? "not reasonably likely" | **residual:** conditional + revocable ? publish/leak
the source later and the link can snap back (forward-looking re-test duty) |
125 | **TM5 re-ID adversary** | source unpublished + reset timeline ? no back-match against
a public reference | ? becomes live if a Fork-A source is ever published ? **and the committed
per-frame trajectory IS a fingerprint** (de Montjoye 4-point re-ID), NOT just ~5 scalars
(corrected §4c). The protection is the unpublished source, not minimization |
126 | **TM6 accidental plaintext commit** | required CI leak-scan rejects MD5/known-
name/non-surrogate/plaintext-SOPS | if the required check is disabled, a leak lands |
127
128 ---
129
130 ## 6. Honest limits ? what a green suite does NOT prove
131
132 - Green proves only that the **deterministic post-trajectory transforms** are unchanged
for the **frozen cases**. It does **not** cover: the WASB/TrackNet detector,
`_probe_fps_width/cv2 frame-peek, the AI/provider handoff, chunking/re-encode, the ffmpeg
stitch, DB persistence/migrations, or the API filter. *A refactor that breaks decoding or the
detector passes these gates green.* The owner-box `regression.py` DB run remains the only full-
chain check.
133 - Characterization = **behaviour-locking, not correctness** ? it freezes current
behaviour *including current bugs*. "Passing" = "behaviour unchanged", never "behaviour
correct". This caveat must ride in the **test pass/fail message**, not only the docs.
134 - **Synthetic-tier floors = plumbing correctness, NOT field quality.** Never cite
synthetic F1 in `QUALITY_ITERATIONS.md` / the paper as rally-detection performance (tag `kind:
synthetic`).
135 - Tier-0 public coverage is **shuttle-only** ; the person/audio fusion ?F1 and exact
Gate-A values need Tier-1 (skipped on no-key clones).
136 - Data is **not "anonymous"*** absolutely (EDPS v. SRB) ? relative, conditional,
revocable, forward-looking.
137 - The committed Tier-0 stream is the **de-attributed per-frame `trajectory.csv`**, NOT
~5 scalars ? a per-frame trajectory is a de Montjoye fingerprint; the only protection is the
**unpublished source** (§4c ?, §5 TM5).
138 - **Time-origin reset is metric-invariant within `_FLOAT_TOL`, not byte-exact for every
offset.** The genuine (raw) reset-vs-no-reset score delta is at the float-error floor (~3.3e-7);
it is **bit-identical for whole-second offsets**, but because 6dp-rounding is not additive
(`round6(a)-round6(b) ? round6(a-b)`) the *serialized* 6dp boundary error can differ by one 6dp
ULP (1e-6) on a sub-second offset ? the serialization grid, not a metric divergence. (Audit fix
#2 rounds `t0`+labels to 6dp before the shift to keep the residual ? the grid step.)
139 - Encryption gives **no secrecy from a running contributor** ; protobuf gives **no
secrecy at all**.
140 - De-attribution does **not** cure provenance/licensing ? it sits *behind* the per-
record provenance gate + counsel sign-off.
141 - `--refresh-snapshot` can rubber-stamp a real regression; mitigation is **review of the
visible diff** + a separate, deliberate floor-baseline update. **A recorded `--reason` is
REQUIRED** (the CLI errors without one ? audit fix #6).
142
143 ---
144
145 ## 7. Deliverables & progress tracker ? **source of truth**
146
147 Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. One **small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
148
149 | ID | Deliverable | Depends on | Counsel-gated? | Status | PR |
150 |---|-----|-----|-----|-----|---|
151 | **P0** | **Leak remediation + required CI leak-scan.** Scrub real names/paths in
`eval_baselines/heuristic_lovo_n6.json`, `gemini_wrapper_2026-06-10.json`,

```

```

`backend/eval/eval_partial_labels.py`, `tests/test_cleanup_caches.py` ? opaque surrogate keys +
private map; add CI leak-scan (MD5 / known-name / non-surrogate / plaintext-SOPS) as a
**required status check**; decide on git history (PR #77, accept vs `filter-repo`). | ? | No
(owned data hygiene) | ? | ? |
152 | **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper** (also
delivered M2-core + M3 ? see below). `backend/eval/golden_fixtures.py` (`replay_fixture`, schema
dispatch, `Path(__file__)`-relative) + 3 **tool-generated** synthetic fixtures + manifest +
README + `.gitattributes` LF pin + `tests/test_golden_fixtures.py`. CONTROL-only (no numpy
execution on the replay path); cross-OS verified. | ? | No (synthetic) | ? |
[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |
153 | **M2** | **Freeze/refresh extractor.** Core (`freeze_synthetic` / `refresh_snapshot` /
`--check` CLI + idempotence test) shipped in M1 (#189); the `--license`/`--minors-
free`/`--owned` **refusal gate** shipped with P1 (this PR). | M1 | No | ? |
[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) · this PR |
154 | **M3** | **Characterization gate** ? `tests/test_golden_fixtures.py`: parse-compared
(exact-int / approx-float 1e-6), **CONTROL-only**, positive no-person/audio assertion,
perturbation + idempotence. **Shipped in M1 (#189).** (Windowing/label fingerprint guard + row-
count cap ? fold into M4.) | M1 | No | ? | [#189](https://github.com/avidullu/badminton-
highlight-indexer/pull/189) |
155 | **MX** | **Cross-OS determinism guard (CI).** Lightweight `golden-crossos` CI job
(ubuntu/windows/macOS) running `--check` + the fixtures test ? locks the cross-OS guarantee
(CONTROL replay is numpy-only, no GPU stack; a *separate fast job*, not full-suite×3). | M1 | No
| ? | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) |
156 | **M4** | **Quality-floor gate + first `regression_baseline.json`.**
`tests/test_golden_quality_floor.py`; create `eval_baselines/regression_baseline.json` with
**windowing fingerprint**; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for
runtime. | M1, M2 | No | ? | ? |
157 | **P1** | **De-attribution + minimization** ? `golden_fixtures.freeze_real_fixture()`
(extends the M1 module, not a separate script): **opaque random `fx_hex` surrogate slug**,
metadata strip, metric-invariant **time-origin reset**, strict GT loader, **positive no-
person/audio/MD5/source-path assertion**, manifest non-clobber; `--freeze-real` CLI with the
**refusal gate** (exit 2 unless owned + minors-free + license). Tooling-only, 12 synthetic
tests; **no real data committed**. | M1 | No (tooling) | ? | this PR |
158 | **P2** | **Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the
researched IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source ·
minors excluded · biometric/person streams excluded · de-attributed). Owner's informed risk
acceptance ? **NOT a retained-counsel opinion**; optional future counsel confirmation per §8. |
P1 | Owner-accepted | ? | ? |
159 | **P3** | **Tier-0 REAL subset.** Commit de-attributed, minors/biometric-excluded,
owned Fork-A real fixtures (unblocked by P2). Hard-gated **in code** by P1's refusal gate (owned
+ minors-free + license required) ? needs the owner's per-video clearance list. | P1, P2 |
conditions §3 | ? | ? |
160 | **O1** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1`
proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
161 | **O2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
162 | **DOC** | **Discoverability cross-links** ? added the `eval_baselines/fixtures/` row
to `CODE_MAP §0` + a `golden_fixtures.py` entry to `CODE_MAP §2.3`; cross-linked
`GOLDEN_DATA_SHARING.md` (committed subset ? private OneDrive corpus). README indexed in #186;
memory `current-state` kept live. (EVALUATION/QUALITY_ITERATIONS cross-links **deferred** ? that
doc surface is hot in the concurrent quality track; avoid collision.) | M1 | No | ? | this PR
|
163
164 | **Must-fix-before-any-public-commit (red-team), folded into the rows above:** ? numeric-
tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5/MX ?); ? scrub names in
**all** files + history decision (P0); ? leak-scan a **required** check, hooks are convenience
only (P0); ? person-optional loader + **positive** no-person assertion, don't trust the silent
`0.0` default (P1 ?); ? random surrogate id not HMAC-over-roster (P1 ?); ? quantize/drop
fps/frame_width (P1 ? fps/fw retained as scale constants; quantize is a P3 option); ? document
Tier-0 omits the fusion path (DOC); ? counsel + Fork-A + minors/biometric exclusion as hard
preconditions (P2 owner-accepted; enforced in code by P1's refusal gate ?).
165
166 | ### Progress log (quantitative ? coverage must hold/improve, CI incl. the 3-OS cross-OS

```

```

job green)
167 | PR | Deliverable(s) | Suite | Coverage | Notes |
168 |---|-----|-----|-----|-----|
169 | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) | M1 +
M2-core + M3 | 746 pass / 7 skip | 80.41% | synthetic Tier-0 + CONTROL replay +
characterization; cross-OS proven (CI Linux replayed Windows-frozen fixtures) |
170 | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) | MX +
tracker/legal reconcile | full suite green | 80.41% | `golden-crossos` CI job green on
ubuntu+windows+macos ? cross-OS **enforced**; legal ? owner-accepted |
171 | _this PR_ | P1 + M2 refusal gate | 759 pass / 7 skip | **80.49%** (+0.08) |
`freeze_real_fixture` + refusal gate; **+13 tests**; tooling-only, no real data committed |
172 | _this PR (audit-hardening)_ | adversarial-audit fixes #17#17 | 784 pass / 7 skip |
**80.71%** (+0.22) | strict-slug `label` + dropped `source_note` (#1); 6dp-rounded reset within
`_FLOAT_TOL` (#2); temp-dir-then-promote + negative-label refusal (#3); exact-membership source-
path scan (#4); key-set-equality in `compare_expected` (#5); `--reason` mandatory (#6);
independent schema guard (#7); `max_win` recorded-not-applied note (#8); numpy-execution wording
(#9); +failure-semantics & by-kind tests (#10/#11); `test_golden_real_fixtures.py` added to
`golden-crossos` CI (#12); doc reconciled to shipped reality incl. per-frame-CSV honesty + risk
re-confirm (#13?#17). Synthetic `--check` byte-identical (unchanged). |
173
174 ---
175
176 ## 8. Open owner / counsel questions
177
178 **Counsel questions ? documented basis (the owner accepted the researched risk on
2026-06-16; see P2). These remain the record, and a retained-counsel confirmation is
optional/recommended before scaling beyond a small owned, minors-free, de-attributed subset:**
179 1. Does the EU/UK **sui-generis database right** subsist in our self-derived
shuttle/rally streams on the **Sportradar** "obtained vs created" reasoning, and does
redistributing a de-attributed Tier-0 subset cause **CV-Online** harm?
180 2. Post **Thomson Reuters v. Ross** + USCO-2025, is **US fair use for the extraction
step **defensible if the dataset could compete in a "data/analytics" market ? does publishing
only numbers (never footage) matter?
181 3. India: are anti-scraping ToS enforceable; **IT Act s.43/s.66** exposure for any 3rd-
party ingestion; does pending **ANI v. OpenAI** (~Apr 2026) move it? Confirm **DPDP s.9/Rule 10**
and whether per-player tracking is prohibited s.9(3) monitoring of a minor.
182 4. Sign-off on the **final committed Tier-0 subset + provenance log**; is non-
attributability-via-private-source a defensible posture to assert given its
conditional/revocable nature?
183
184 **Owner (strategic):**
185 5. Publishing the Fork-A subset **forfeits the trade-secret moat** (Indian breach-of-
confidence framing) ? weighed against community/benchmark/paper aims. Your call.
186 6. **Forward-looking commitment:** do not publish a highlight reel of the **same** source
whose fixtures are committed (without re-clearing), and accept a periodic re-assessment as the
public-video environment grows.
187 7. **Scope of first public commit:** synthetic-only (zero provenance risk, recommended)
vs a counsel-cleared real subset.
188 8. **Tier-1 mechanism:** out-of-repo+token (recommended) vs SOPS+age in-repo. And: is
protobuf (01) wanted, or is canonical-JSON enough?
189
190 ---
191
192 ## 9. Definition of done
193
194 The project is **DONE** (flip `Status ? DONE`, archive) when:
195 - [ ] **P0, M1?M4, P1, DOC** are ? and merged.
196 - [ ] A clean `git clone` on **win + ubuntu + mac** runs `python run_all_tests.py` and
the **Tier-0 gates pass green with no key / no video / no GPU / no DB**.
197 - [ ] The **CI leak-scan is a required status check** and passes (no MD5 / real-name /
non-surrogate / plaintext-SOPS in the tree).
198 - [ ] Characterization + quality-floor gates are **parse-compared, CONTROL-only,
windowing-fingerprinted**, and demonstrably catch a seeded behaviour change.
199 - [ ] `docs/CODE_MAP.md` §0 + TEST MAP updated; this doc cross-linked; `NEXT_STEPS.md` +
memory `current-state` point here.

```

```

200     - [ ] Either the public commit is synthetic-only, or (P2 owner-accepted ? +
P3) for a real-data commit, under the S3 conditions enforced in code by P1's refusal gate.
201     - [ ] S6 honest-limits caveat rides in the test pass/fail messages, not only the docs.
202
203     *Optional 01/02 may remain ? without blocking DONE (they're earned, not assumed).*
204
205     ---
206
207     ## 10. References
208
209     Internal code anchors (verified): `backend/eval/{regression.py, fusion_golden.py, fusion_compare.py, ablation.py, experiment.py, metrics.py, windowing.py, distill_local.py, gt_loader.py, eval_stats.py}` · `backend/pipeline/detectors/{base.py, tracknet_runner.py, rally_gate.py}` · `backend/utils/hashing.py::compute_video_id` · `eval_baselines/{heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json, experiment_leaderboard.json}` · `tests/test_ablation.py::test_litmus_reproduces_gate_a` · `.github/workflows/ci.yml`.
210     Internal docs: `GOLDEN_DATA_SHARING.md` (#175), `EVALUATION.md`, `EXPERIMENT_HARNESS.md`, `QUALITY_ITERATIONS.md`, `MULTI_SIGNAL_FUSION_PLAN.md`, `CODE_MAP.md` S0`.
211     Legal authorities: see S3 (full URLs in the research workflow artifact). *This section is researched general information, not legal advice.*
212
213     ---
214
215     ### Changelog
216     - 2026-06-16 ? Initial draft synthesized from two multi-agent workflows (mechanics design + 2 critiques; legal memo IN/US/EU + SEALED-FIXTURES architecture + red-team). Verified the live name-leak and the BLAS-determinism cause against code. Status: DRAFT, owner-gated.
217

```

11. assistant (2026-06-19T17:18:25.386Z)

Now let me gather cross-reference evidence. I'll grep for each filename across the key live docs and the whole repo, and check archives for already-archived copies.

12. user (2026-06-19T17:18:27.408Z)

```

60:- Substantial design/project work gets a tracked project doc following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` S7,
`docs/GOLDEN_REGRESSION_FIXTURES.md`.

```

13. user (2026-06-19T17:18:29.295Z)

```

20:2. Artifacts land in the shared cloud folder `C:\Users\avidu\OneDrive\rally-annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
[`GOLDEN_DATA_SHARING.md`](GOLDEN_DATA_SHARING.md); tracked in ##175.

```

14. user (2026-06-19T17:18:30.777Z)

```

12:[Omitted long matching line]
21:- [GOLDEN_DATA_SHARING.md](GOLDEN_DATA_SHARING.md) ? How the GPU box and CPU dev/agent
sessions share golden artifacts (trajectory CSVs, `*_golden_features.json`) via a OneDrive
folder; the GPU-extract ? CPU-litmus split. See ##175.
24:- [PROJECT_DOC_TEMPLATE.md](PROJECT_DOC_TEMPLATE.md) ? Reusable skeleton for a tracked
project doc: status header · decisions-locked · progress tracker (??/?) · definition-of-done
· archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` S7
and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.

```

15. assistant (2026-06-19T17:18:35.752Z)

Let me check the README.md line 12 (omitted) and grep the whole repo for inbound references plus check archives for already-archived copies.

16. user (2026-06-19T17:18:37.229Z)

1 - ROORA_LABELING_GUIDELINES.md ? Latest (v8) vendor labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-driven court-calibration / singles-doubles additions marked `? ADDED (fusion)`.

2 - RALLY_QUALITY_RESEARCH.md ? **Foundation doc** for closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps, the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-class, golden-corpus + active learning, Gemini-agreement as a **shadow** signal), an ***adversarially-verified, cited external-research synthesis*** (TAL/TAS boundary heads, racket-sports rally-state cues, small-data, eval rigor), a ***prioritized cheap?durable roadmap with falsifiable success criteria***, and (§7) a ***tracked project plan*** ? work-item checklist, sequencing, the single quality ***number*** attributed to the effort, and a ***definition of done*** (project completes when all tiers land + the number is recorded). The plan that the QUALITY_ITERATIONS.md log executes against; ties together EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS / OWNED_MODEL_IMPLEMENTATION_PLAN.

3 - RALLY_DETECTION_GAP_CLOSURE.md ? ***Tracked execution doc (2026-06-18)*** for closing the remaining rally-detection accuracy gap ***per video, before launch***, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it: boundaries are tight (R2/R3/R4 holds) ? the gap is ***detection***: (G1) recall (local & Gemini both miss ~half on hard clips, and **different** rallies ? complementary), (G2) precision (local over-fires), (G3) ***rally-END over-extension on the quick-return-after-point*** (shuttle still moving ? needs a **player-state** signal). Levers (owner-gated): player-kinematics + shuttle-in-hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally\_gate` Gate A wiring, complementary fusion. ***Peer*** of `RALLY\_QUALITY\_RESEARCH.md` (execution tracker, not research); §7 is the source of truth.

4 - PROMINENT_COURT_DETECTION.md ? ***Design + experiment (2026-06-19, default-OFF, owner-gated flip)*** for the ***multicourt G2 over-fire***: identify the ***prominent (foreground) court*** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` ***before*** windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? ***pseudo-3D play-volume*** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion\_features.point\_in\_polygon` + `FusionConfig.court\_polygon`. §8 progress tracker is the source of truth.

5 - RALLY_DETECTION_QUALITY_REPORT.md ? ***Technical-paper checkpoint (2026-06-17)*** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.091±0.323 at n=11, p=0.002; merge-rate 0.694±0.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY\_QUALITY\_RESEARCH.md` + `QUALITY\_ITERATIONS.md`.

6 - EVALUATION.md ? How we measure and improve quality.

7 - EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with ***zero production regression*** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.

8 - QUALITY_ITERATIONS.md ? ***Living log*** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.

9 - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a ***signal-fusion framework*** for rally detection ? producers emit confidence-scored ***signals*** at three temporal ***scopes*** (frame detectors / window analyzers / session analyzers), a learned ***scorer*** weights them, and an auditable ***report-first reconciliation*** pass lints the decision against the evidence. The spine: ***no special-casing in the decision path*** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT\_HARNESS.md` and `MULTI\_SIGNAL\_FUSION\_PLAN.md` (ADR-007 modular rally layer).

10 - PERSONALIZATION_PLAN.md ? Design (owner-adopted ***hybrid***; Phase-0 landing) for the ***per-user personalization feature on a generalization-first baseline***: a thin per-user ***tuning*** layer gated by the honest small-N statistics (a ***min-N**

promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *a tool that gets better the more you help it*). Records the **locked owner decisions** (identity, storage, `min\_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

11

12 - GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started):** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **\$7 progress tracker + definition-of-done**. Extends GOLDEN_DATA_SHARING.md.

13

14 ## Product / platform plans

15 - DESKTOP_APP_PLAN.md ? **Tracked scoping doc (2026-06-18, owner-gated, `[design]` ? NOTHING built):** package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? \$7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

16

17 ## Practical / ongoing

18 - TRACKNET_WSL_SETUP.md
19 - Golden set docs (`GOLDEN\_SET`)
20 - Evaluation harness and calibration drivers (in `backend/eval/`)
21 - GOLDEN_DATA_SHARING.md ? How the GPU box and CPU dev/agent sessions share golden artifacts (trajectory CSVs, `\_golden\_features.json`) via a OneDrive folder; the GPU-extract ? CPU-litmus split. See **\$175**.

22

23 ## Conventions & templates

24 - PROJECT_DOC_TEMPLATE.md ? Reusable skeleton for a **tracked project doc**: status header · decisions-locked · progress tracker (???) · definition-of-done · archived on completion. Standardizes the emergent pattern in `RALLY\_QUALITY\_RESEARCH.md` §7 and `GOLDEN\_REGRESSION\_FIXTURES.md`. Use for any substantial design/project work.

25

26 ## Historical documents

27 Completed plans, old research, ADRs, and checkpoints live in archives/.

28

29 - CODE_AUDIT_AND_TEST_HARDENING.md ? **COMPLETED** (2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing / contract / architectural audit, 626-test infrastructure, positive test plans for the 5 identified issues, multi-layer regression protocol (including ablation/experiment layer), and pre/post-change verification checklist. See `docs/archives/CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md` for the final archived state. (Top-level is now pointer stub per review; full content in archives/.)

30

31 See `archives/README.md` for the umbrella policy and `archives/past\_projects/README.md` for classification of small tech debt / starter projects.

17. user (2026-06-19T17:18:38.021Z)

No files found

18. user (2026-06-19T17:18:38.057Z)

No files found

19. assistant (2026-06-19T17:18:40.863Z)

No archived copies of any GOLDEN* doc exist. Now let me get full inbound reference counts across the whole repo for each filename.

20. user (2026-06-19T17:18:41.986Z)

Found 5 files

docs\BACKLOG.md
docs\GOLDEN_SET_VENDORING.md
docs\archives\decisions\DECISIONS.md
run_regression.py
backend\eval\regression.py

21. user (2026-06-19T17:18:44.842Z)

Found 3 files

docs\BACKLOG.md
docs\VIDEO_RECORDING_GUIDELINES.md
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md

22. user (2026-06-19T17:18:44.900Z)

Found 13 files

docs\RALLY_QUALITY_RESEARCH.md
docs\MULTI_SIGNAL_FUSION_PLAN.md
docs\BACKLOG.md
docs\VIDEO_RECORDING_GUIDELINES.md
docs\ROADMAP.md
docs\ROORA_LABELING_GUIDELINES.md
docs\COMMERCIALIZATION.md
docs\EXPERIMENT_HARNESS.md
docs\archives\decisions\DECISIONS.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\GOLDEN_SET_IMPLEMENTATION.md
backend\annotations__init__.py
backend\annotations\base.py

23. user (2026-06-19T17:18:48.058Z)

Found 5 files

docs\RALLY_DETECTION_GAP_CLOSURE.md
docs\QUALITY_ITERATIONS.md
backend\eval\rally_seg_eval.py
tests\test_rally_seg_eval.py
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\README.md

24. user (2026-06-19T17:18:49.351Z)

Found 3 files

docs\README.md
docs\NEXT_STEPS.md
docs\GOLDEN_REGRESSION_FIXTURES.md

25. user (2026-06-19T17:18:49.379Z)

Found 12 files

docs\README.md
docs\GOLDEN_VIDEOS.md
docs\CODE_MAP.md
CLAUDE.md
backend\eval\golden_real_fixtures.py

```
backend\eval\golden_fixtures.py
eval_baselines\fixtures\README.md
docs\REAL_FOOTAGE_VALIDATION_REPORT.md
docs\GOLDEN_DATA_SHARING.md
.github\workflows\ci.yml
.gitattributes
docs\PROJECT_DOC_TEMPLATE.md
```

26. assistant (2026-06-19T17:18:52.654Z)

I now have enough evidence to make decisions. Let me check NEXT_STEPS.md briefly to confirm whether the GOLDEN_SET_* docs are cited as live next-work (the README "Practical / ongoing" section lists them generically).

27. user (2026-06-19T17:18:54.535Z)

No matches found

28. assistant (2026-06-19T17:19:08.537Z)

Now I have a complete evidence picture. Let me summarize my analysis before calling StructuredOutput:

****GOLDEN_SET_IMPLEMENTATION.md**** ? Status "In progress". Tracker shows GS-1/GS-2/GS-4 landed; GS-3 (HumanVendorProvider) parked pending Roora pilot, GS-5 still pending. This is an implementation tracker with open slices ? LIVE in-progress work. Referenced by code (run_regression.py, backend/eval/regression.py) and GOLDEN_SET_VENDORING.md. KEEP_LIVE.

****GOLDEN_SET_PHASE1_VIDEOS.md**** ? Status "Design v1". A still-intended collection spec; golden-set creation described as iterative/ongoing. Owner-gated design ? LIVE future work. KEEP_LIVE.

****GOLDEN_SET_VENDORING.md**** ? Status "Design v1" + ADR-006. Roora pilot in progress, gold key + GS-3 pending. LIVE design/process doc. KEEP_LIVE.

****GOLDEN_VIDEOS.md**** ? Living registry ("Updated whenever a new video is labeled + ingested"), n=15 growing corpus, last updates 2026-06-17. Actively maintained reference. KEEP_LIVE.

****GOLDEN_DATA_SHARING.md**** ? Status "convention + plan (2026-06-15), Tracked in #175". Cited in NEXT_STEPS.md and README.md as a live workflow. Active convention. KEEP_LIVE.

****GOLDEN_REGRESSION_FIXTURES.md**** ? Status "IN PROGRESS ? owner-gated". §7 tracker shows M1/M2/M3/MX/P1/P2/DOC done but P0, M4, P3, O1, O2 still ?. DoD not met. Cited as live tracked-doc precedent in CLAUDE.md and README. Clearly LIVE in-progress. KEEP_LIVE.

All six are non-terminal. None has an archived copy.

29. user (2026-06-19T17:19:39.307Z)

Structured output provided successfully

30. assistant (2026-06-19T17:19:46.981Z)

All six docs in my batch are non-terminal and should stay live. Summary:

- ****GOLDEN_SET_IMPLEMENTATION.md**** ? KEEP_LIVE. "In progress"; GS-3/GS-5 slices unshipped; referenced by live code.
- ****GOLDEN_SET_PHASE1_VIDEOS.md**** ? KEEP_LIVE. "Design v1"; still-intended capture spec with an open §5 train/eval guardrail TODO.
- ****GOLDEN_SET_VENDORING.md**** ? KEEP_LIVE. "Design v1" / ADR-006; Roora pilot in-flight, gold key + GS-3 pending; 13 inbound refs.
- ****GOLDEN_VIDEOS.md**** ? KEEP_LIVE. Living golden-corpus registry (n=15, updated through 2026-06-17), wired into eval code/tests.
- ****GOLDEN_DATA_SHARING.md**** ? KEEP_LIVE. Active convention (#175), cited as a live workflow in NEXT_STEPS.md.

- ****GOLDEN_REGRESSION_FIXTURES.md**** ? KEEP_LIVE. "IN PROGRESS, owner-gated"; DoD unmet (P0/M4/P3 open); cited as a tracked-doc precedent in CLAUDE.md.

None has an existing archived copy, so no DEDUPE cases. No contradiction traps (no doc claims COMPLETED while cited as live), so no ASK_OWNER cases.